Date:- 13/2/26

Ex-No:- 05

VERIFICATION OF NORTON'S THEOREM.

Aim:-

To verify Norton's theorem practically and theoretically for the given DC circuit.

APPARATUS REQUIRED:-

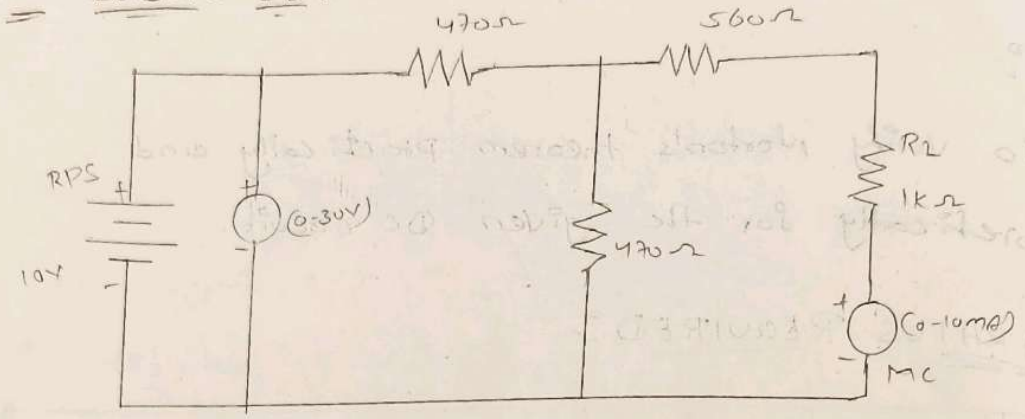
S.No	APPARATUS	SPECIFICATION	QUANTITY
1	Regulated power supply	(0-30) V	1
2.	Voltmeter	(0-30V) MC	1
3.	Ammeter	(0-10mA) MC	1
4.	Resistors	470Ω, 680Ω, 1kΩ.	2, 1, 1.
5.	Bread board	---	1
6.	Multimeter	---	1

THEORY:-

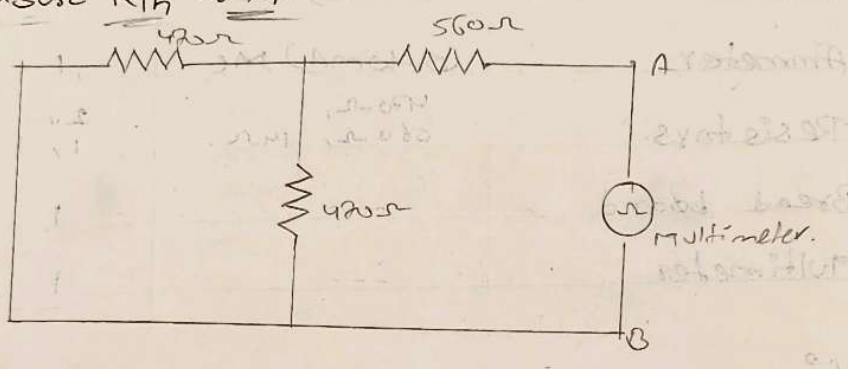
NORTON'S THEOREM:-

Norton's theorem states that, "a linear two terminal circuit can be replaced by an equivalent circuit consisting of a current source, $I_N = (I_{SC})$ in parallel with a resistor $R_N = R_{Th}$, where $I_N = I_{SC}$ is the short-circuit current through the load resistor and R_N is the equivalent resistance at the load terminals when the independent source and turned off

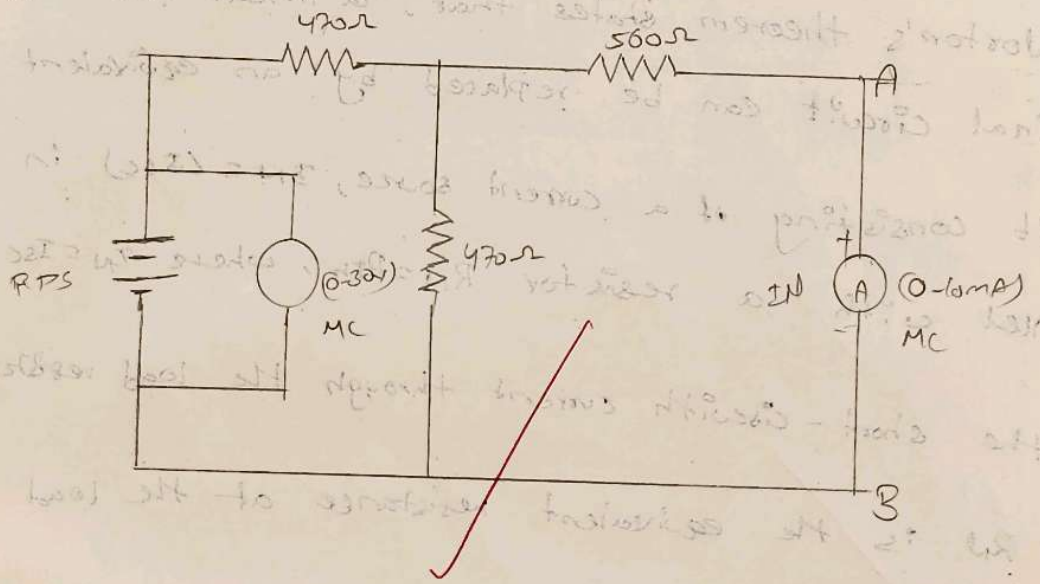
To measure I_L :-



TO measure R_{Th} or R_N :-



TO measure I_N or I_{SC} :-



Norton's current, I_N (or) I_{sc} :-

It is the short-circuit current through the load terminals, i.e, $I_N = I_{sc}$.

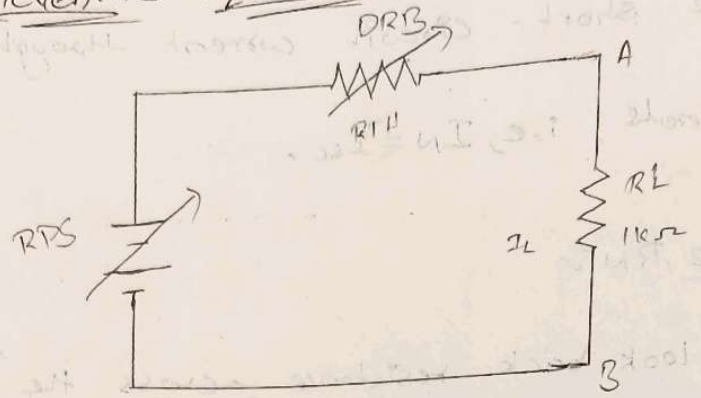
Norton's Resistance, R_N :-

It is the look-back resistance across the load terminals when all the sources are replaced by their internal resistance. An ideal voltage source is replaced by short-circuiting as it's internal resistance is zero. An ideal current source is replaced by open-circuiting as it's internal resistance is infinity.

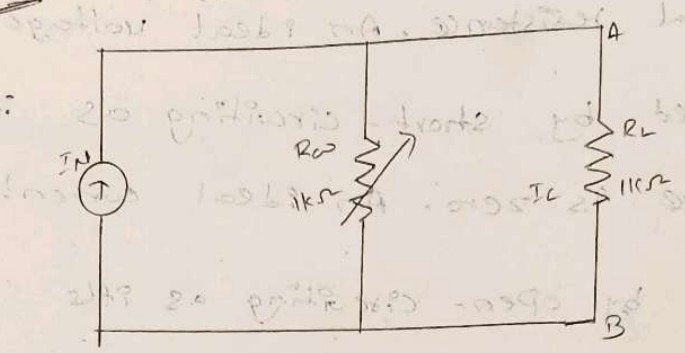
PROCEDURE :-

1. Make the connections as per the circuit diagram 1.
2. vary the RPS and set an input voltage of 10V.
3. Note down the voltmeter reading (V_1) and ammeter reading (I_L) in tabular column 1.
4. switch off the supply and make connections for circuit diagram 2.
5. Measure the thevenin's resistance R_{th} = Norton's resistance R_N.

Thevenin's equivalent circuit:



Norton's equivalent circuit:



Tabular column 1:

To measure I_L:

Vi (volts)	I_L (amps)
10V	275 mA

Tabular column: 2

To measure R_{th} or R_N

Vi (volts)	R_{th} (Ω)
10V	74Ω

6. Switch off supply and make connections for circuit diagram 3.

7. set an input voltage of 10V in the RPS and note down the voltmeter readings V_1 and $V_{TH} = V_{OC}$ in Tabular column 3.

8. switch off the supply and make connections for circuit diagram 4.

9. set an input voltage of 10V in the RPS and note down the voltmeter reading V_1 and ammeter reading $I_N = I_{SC}$ in tabular column 4.

10. Draw the thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagrams 5 & 6 respectively.

11. calculate the I_L value using the formula thevenin's theorem

$$I_L = V_{th} / (R_{th} + R_L)$$

Norton's theorem

$$I_L = I_N * R_N / (R_N + R_L)$$

Tabular column: 3

To measure I_N (or) I_{sc}

V_i (volts)	I_N (mA)
10V	6.27 mA

MODEL CALCULATION:-

Practical value of I_L (from tabulation 1) = 2.3 mA

Verification of Norton's theorem:-

$I_L = I_N * R_N / (R_N + R_L) = 2.43$ mA

10. Theorectically verify the norton's theorem.

S.No	Description	marks Alloted	marks obtained
1.	Experimental set up	20	20
2.	observation/Readings	20	20
3.	Calculations/Result	20	18
4.	Viva	20	18
5.	Record	20	18
6.	Total	100	94

Result :-

Thus, thevenin's and theorem's theorem is verified practically and theoretically.